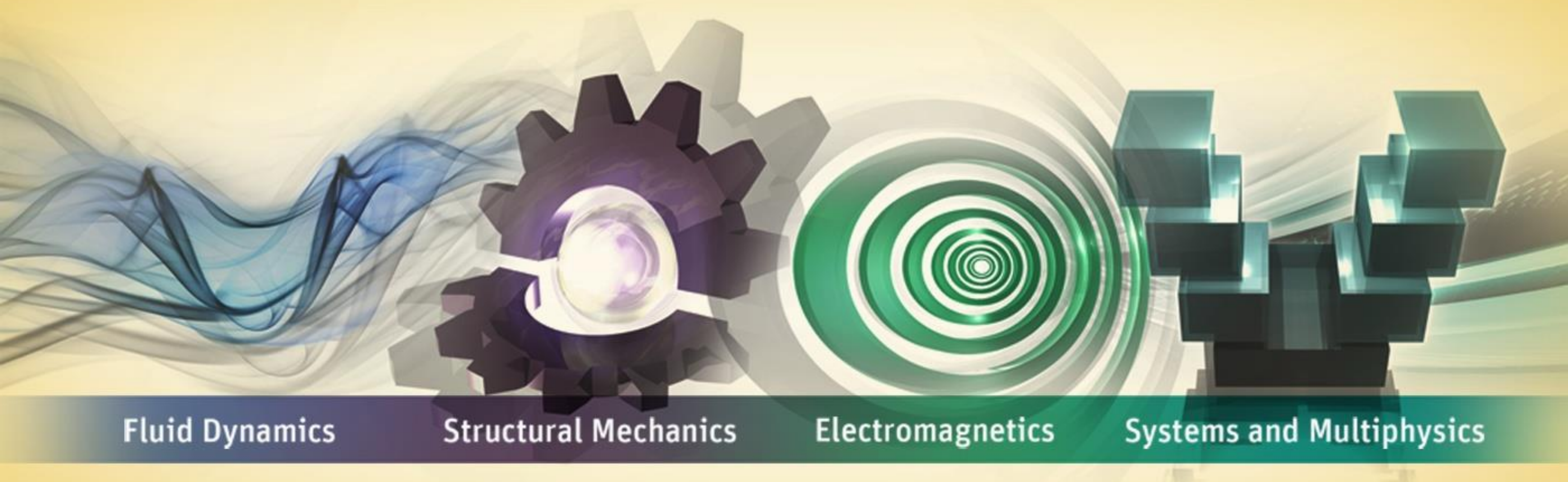
Realize Your Product Promise™

Workshop 2A:
Small vs Large Deflection



ANSYS Mechanical
Introduction to Structural Nonlinearities

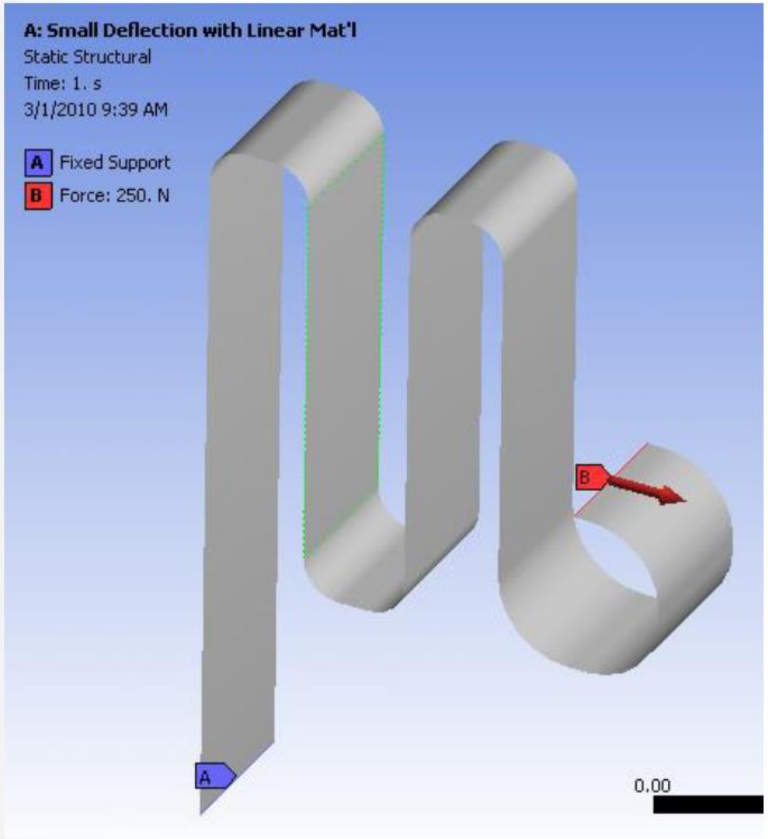
Workshop 2A: Large Deflection

Goal

Compare and contrast results using small deflection theory and large deflection theory on a model with identical loads and boundary conditions.

Model Description

- 3D Spring plate
- Linear steel material
- Meshed with 3D Shell elements
- Fixed support at one end, A
- 250N load at opposite end, B



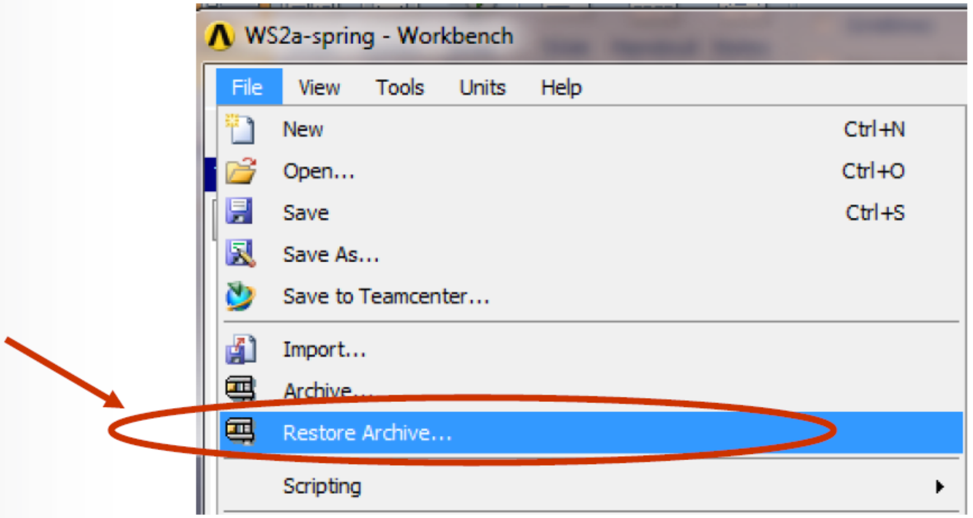
… Workshop 2A: Large Deflection

Steps to Follow:

Start an ANSYS Workbench session.

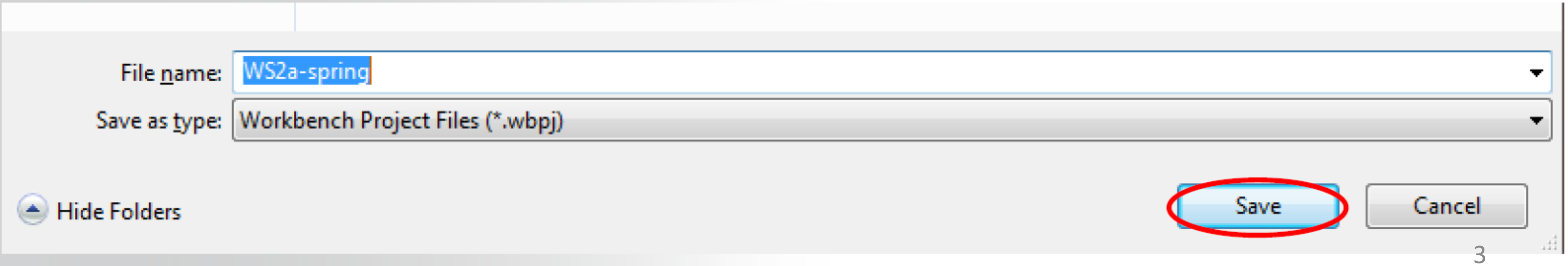
Use File > Restore Archive… browse for existing file "SNL WS2a_spring.wbpz"

- **Location of directory provided by instructor**



Save as

- **File name: "WS2a-spring"**
- **Save as Type: Workbench Project Files (*.wbpj)**

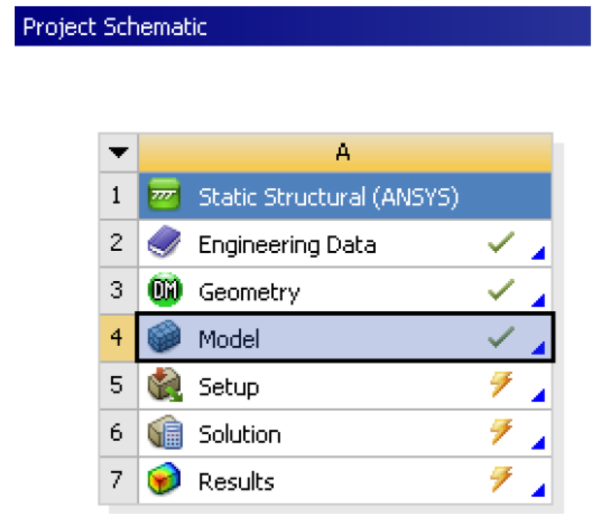


ANSYS

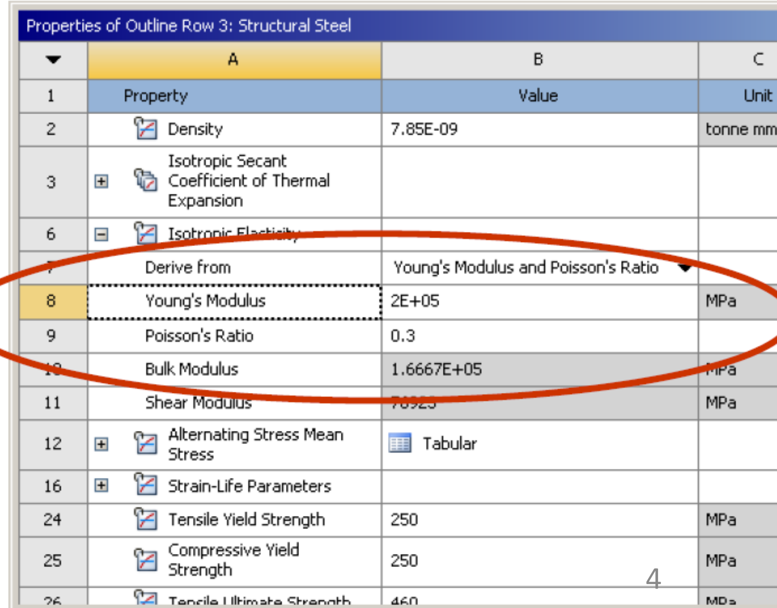
… Workshop 2A: Large Deflection

The Project Schematic should look like the picture to the right.

- **From this Schematic, you can see that the Engineering (material) Data and Geometry have already been defined (green check marks).**
- **It remains to set up and run the FE model in Mechanical**
- **Open the Engineering Data Cell (highlight and double click OR Right Mouse Button (RMB)>Edit) to verify the linear material properties.**
- **To see relevant dialog boxes, it might be necessary to go to Utility Menu > View..**
  - **Click on 'Properties' and 'Outline'**
- **Verify that the units are in Metric(Tonne,mm,…) system. If not, fix this by clicking on…**
  - **Utility Menu > Units > Metric(Tonne, mm,…)**



Small Deflection with Linear Mat'l


Properties of Outline Row 3: Structural Steel

	A	B	C
1	Property	Value	Unit
2	Density	7.85E-09	tonne mm
3	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Young's Modulus and Poisson's Ratio	
8	Young's Modulus	2E+05	MPa
9	Poisson's Ratio	0.3	
10	Bulk Modulus	1.6667E+05	MPa
11	Shear Modulus	76923	MPa
12	Alternating Stress Mean Stress	Tabular	
16	Strain-Life Parameters		
24	Tensile Yield Strength	250	MPa
25	Compressive Yield Strength	250	MPa
26	Tensile Ultimate Strength	460	MPa

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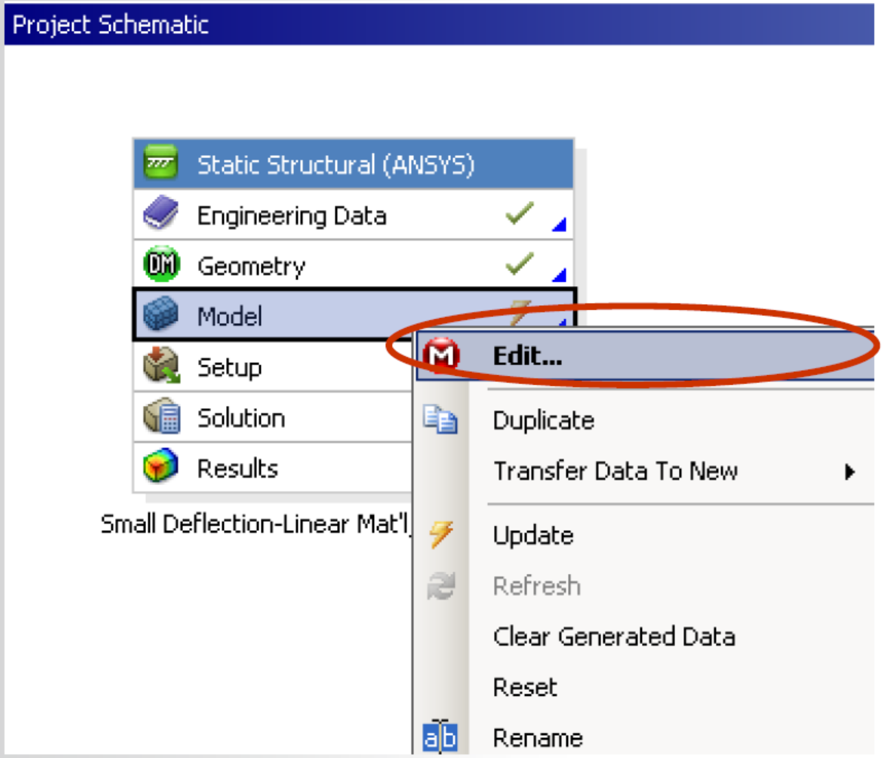
Return to Project Schematic

Double click on the Model Cell to open the FE Model (Mechanical Session) (or RMB=>Edit…)



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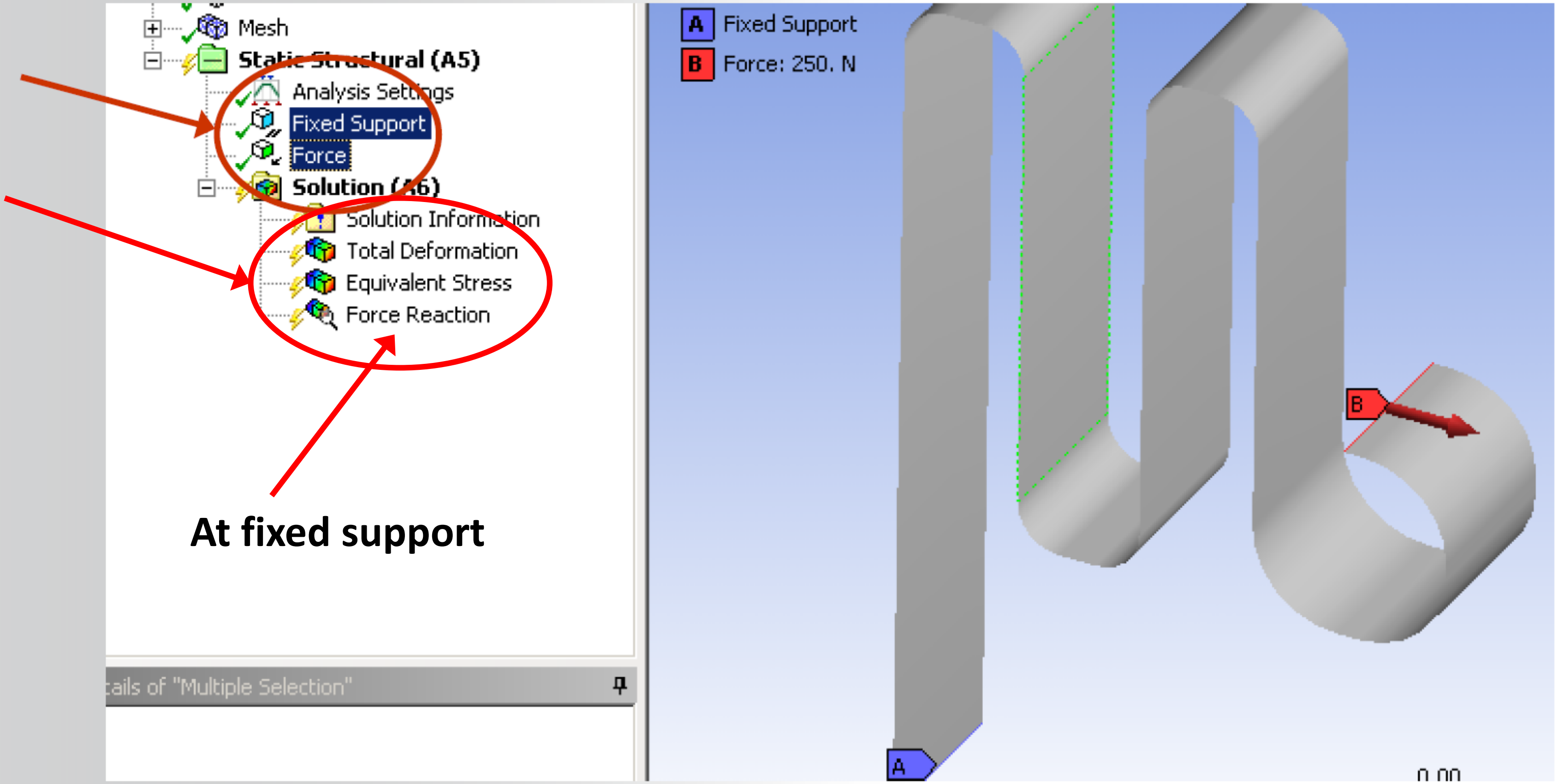
The spring model is already set up with a fixed boundary condition and a force load on the opposite end.

- Highlight the Fixed Support and Force Load to confirm that the model is properly supported, loaded and ready to solve.
- Insert solutions as shown in the picture below.



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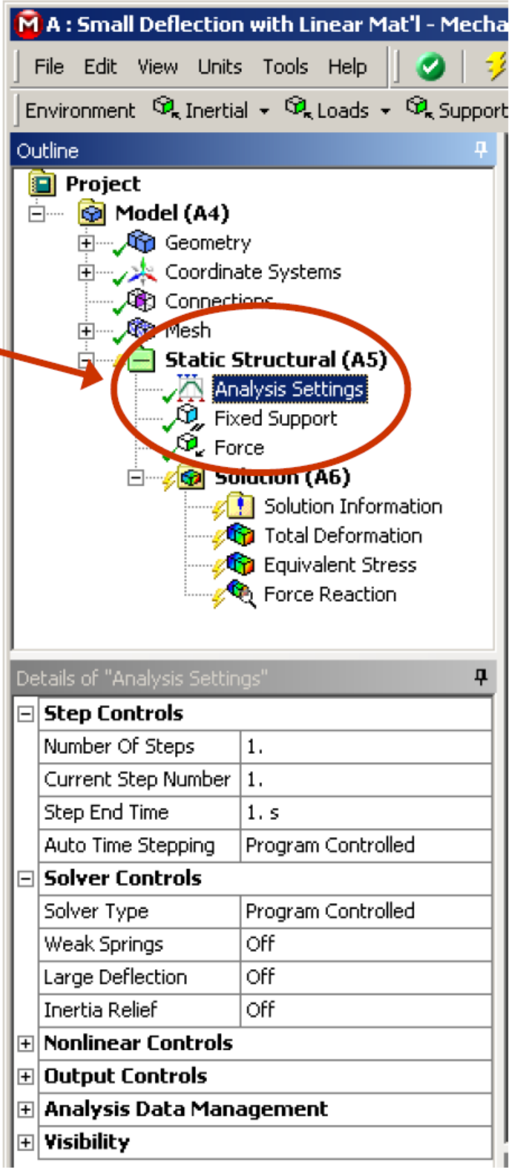
Note the Analysis Settings Specifications:

- Auto Time Stepping = Program Controlled
- Large Deflection = Off

Run the Solution…

Solve



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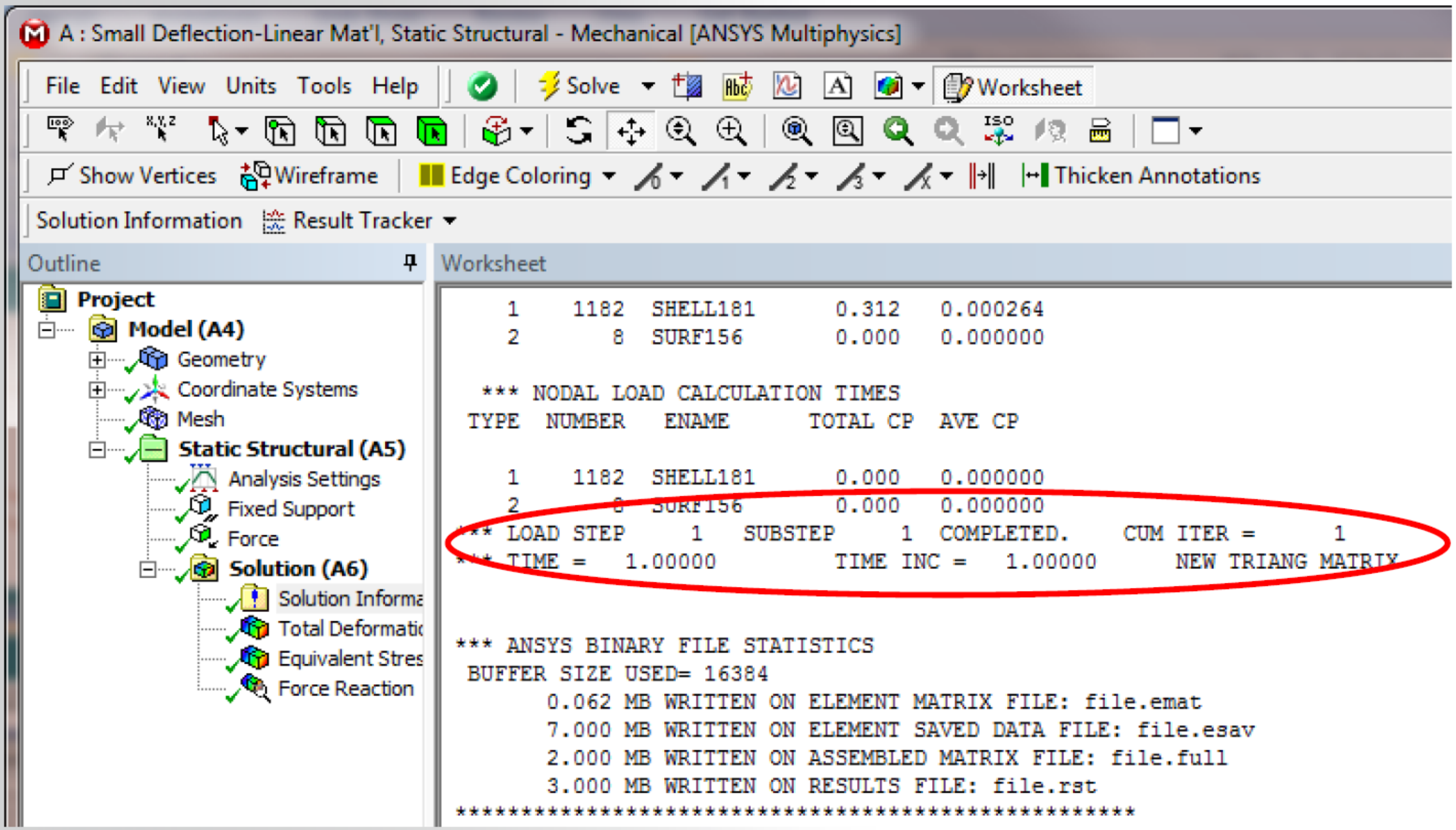
After solution run is complete, highlight the Solution Information folder and scroll to near the bottom of the output.

- **As expected, this solves in one iteration.**



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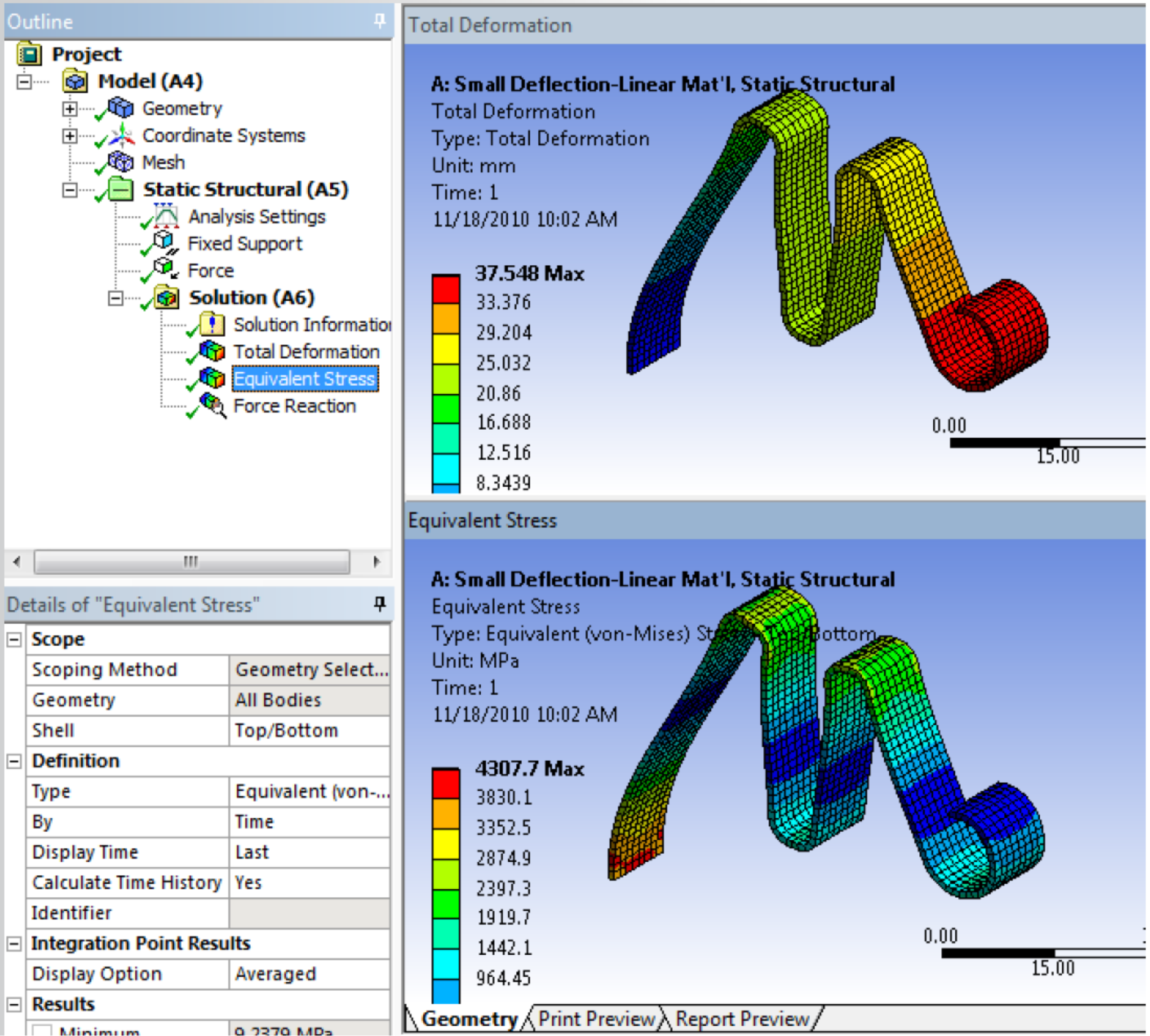
Review the stress and displacement results from this first run.



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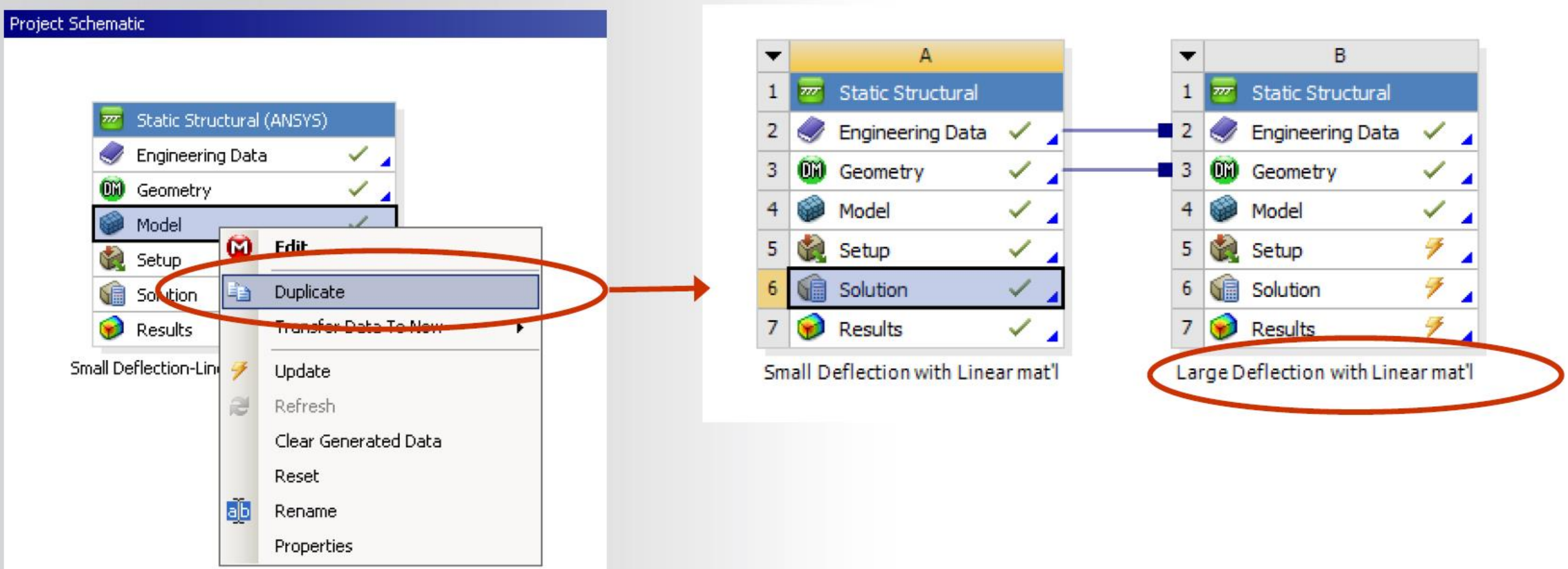
Return to the Project Schematic

- **Highlight the Model Cell, RMB Duplicate…**
- **Rename this new analysis "Large Deflection - Linear Mat'l, Static Structural"**



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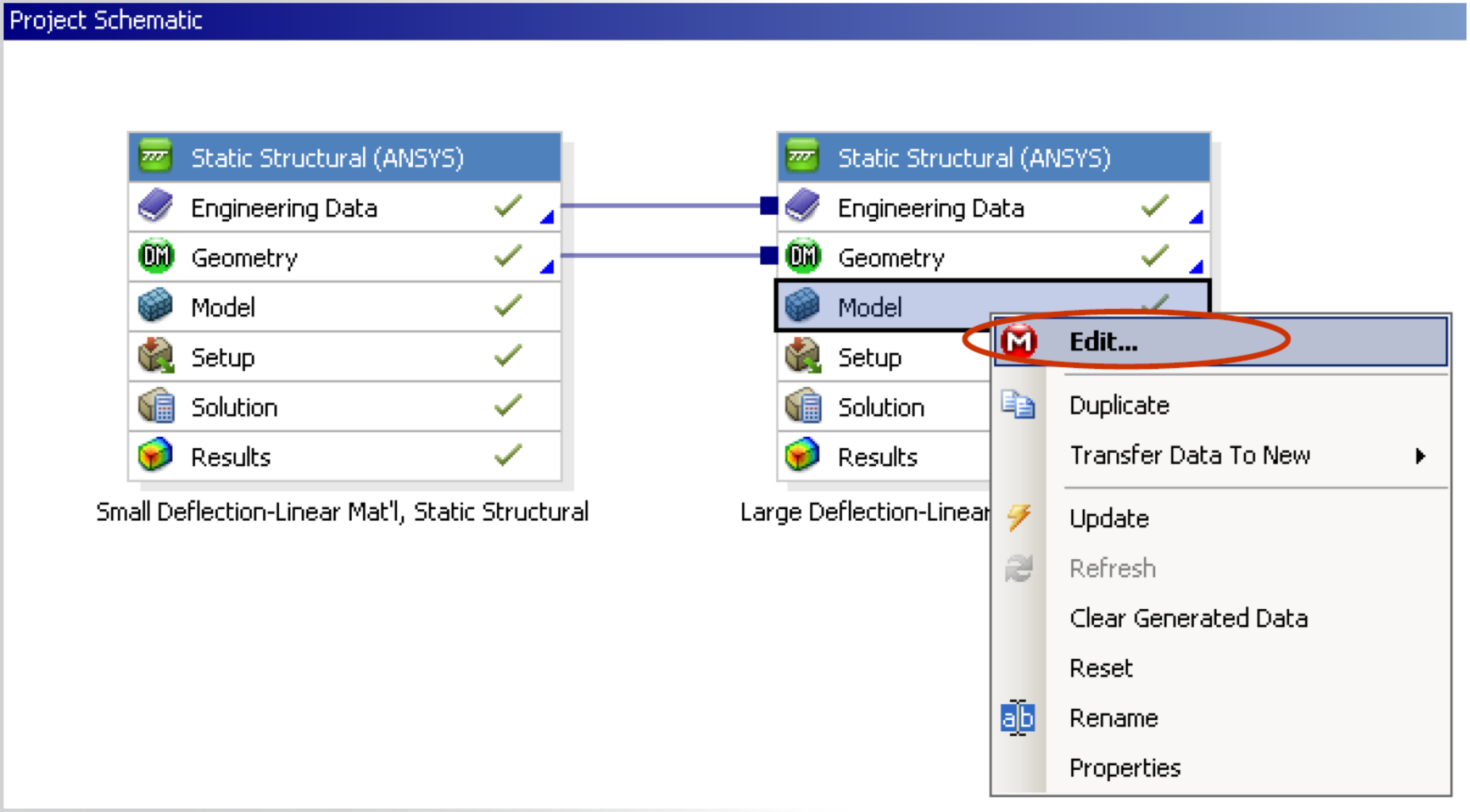
Double click on the Large Deflection Model Cell to open the FE Model (Mechanical Session) (or RMB=>Edit…)



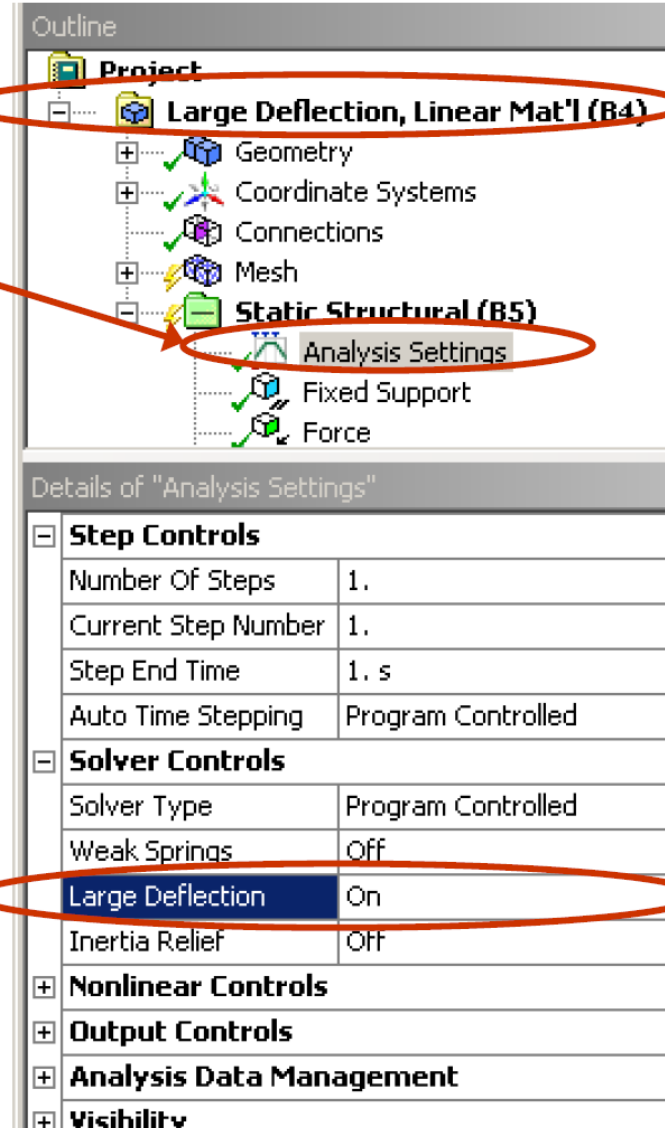
... Workshop 2A: Large Deflection

Highlight the Model folder and change the name to “Large Deflection...

Highlight the Analysis Settings Folder

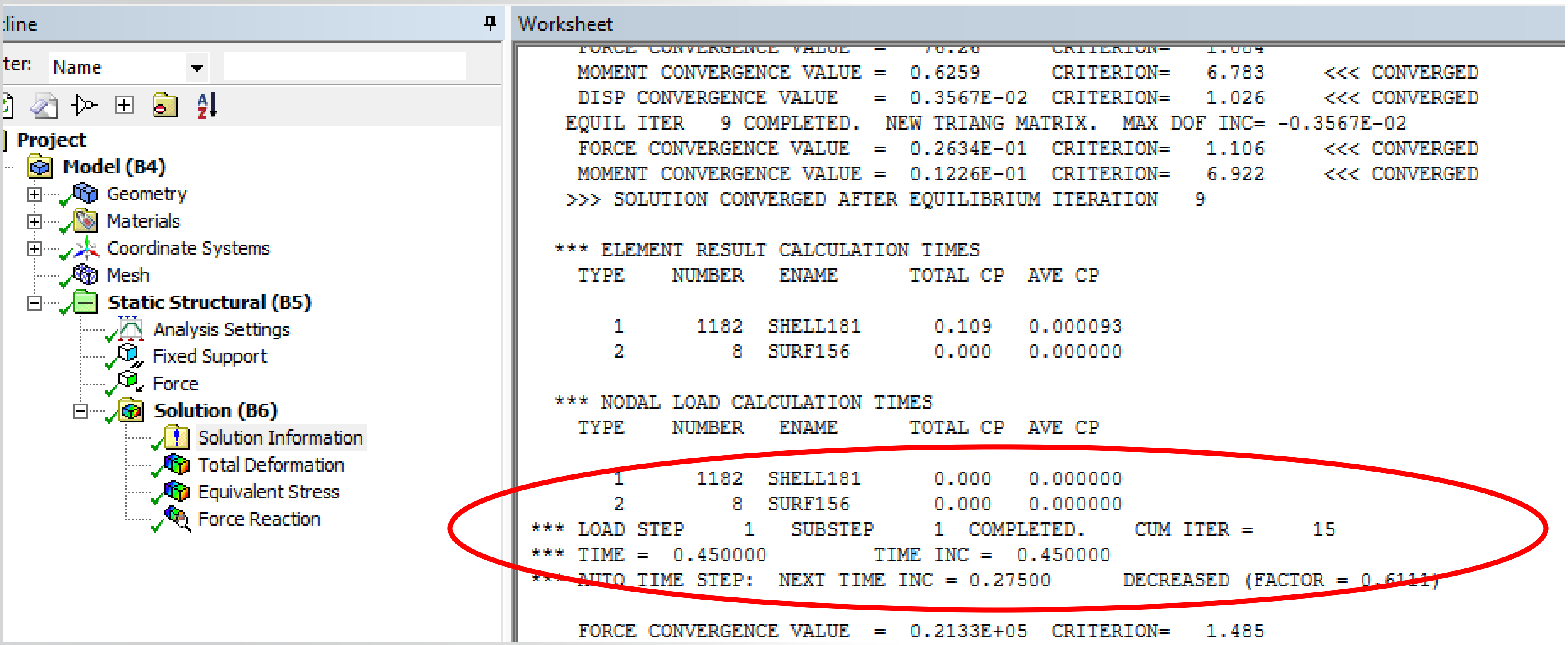
- **Turn on Large Deflection**

Run the new Solution...



… Workshop 2A: Large Deflection

After solution run is complete, open the Solution Information folder and scroll to near the bottom of the output. Note the solution now solves with 15 cumulative iterations made on the stiffness matrix during the run to account for large deflection effects.



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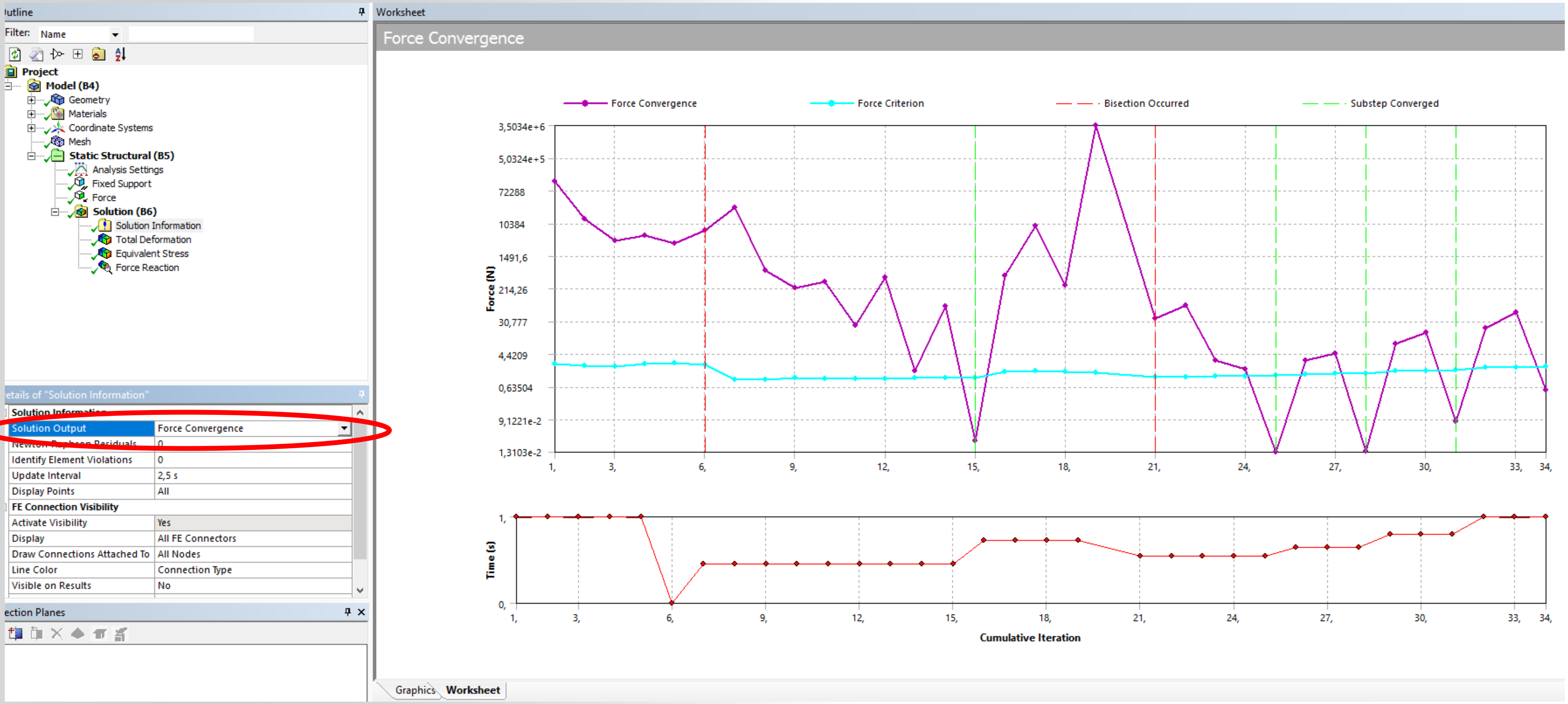
Change Solution Output to Force Convergence to review the Newton-Raphson History.



… Workshop 2A: Large Deflection

- Review the large deflection stress and displacement results and compare with the first run.

- This is an example of how changing shape and stress stiffening effects can have a subtle influence on the results of what would otherwise be a linear problem.

Please keep this model open for use in Workshop 2B